WIRELESS SECURITY

CYS331

INTRODUCTION TO WIRELESS SECURITY

UNIT 1

INTRODUCTION

- **Wireless** is a term used to describe communications in which electromagnetic waves are used for communication purpose.
- **Wireless security** refers to the measures and technologies used to protect wireless networks (Wi-Fi, Bluetooth, cellular networks, etc.) from unauthorized access, cyber threats, and data breaches.
- The first wireless transmitters went on the air in the early 20th century using **radiotelegraphy**.

INTRODUCTION (CONTINUES…)

- **Modulation** made it possible to transmit voices and music via wireless.
- **Modulating** a wave means changing one or more of its fundamental characteristics to encode information.

EXAMPLE OF WIRELESS

1. **Fixed Wireless**

- Refers to a method of delivering high-speed internet or network connectivity using radio signals between fixed locations.
- It connects buildings, homes, or businesses to the internet without the need for cables.
- For example, **Cordless Computer Peripherals** (e.g., **Wireless Mouse, Keyboard, Printers**)

EXAMPLE OF WIRELESS (CONTINUES…)

2. **IR (Infrared) Wireless**

- Is a type of wireless communication that uses infrared light waves to transmit data between devices.
- It is commonly used for short-range, line-of-sight communication and does not penetrate walls or obstacles.
- For example, **TV Remote Controls** and **Medical Devices.**

EXAMPLE OF WIRELESS (CONTINUES…)

3. **Mobile Wireless**

- Refers to wireless communication technology that allows users to stay connected while moving.
- It enables data transmission without the need for a fixed connection, making it ideal for **cell phones**, **tablets**, and other portable devices such as **pagers**.

EXAMPLE OF WIRELESS (CONTINUES…)

4. **Satellite Wireless**

- Is a communication technology that uses satellites in space to transmit data, voice, and video signals over long distances.
- It provides connectivity in areas where traditional cellular or wired networks are unavailable, such as remote regions, oceans, and deserts.
- For example, **Satellite Internet** (e.g., Starlink) and **Global Positioning System (GPS)**.

WIRED VS WIRELESS TECHNOLOGY

	WIRED	WIRELESS
SPEED	Faster & more stable (Ethernet, fiber optic)	Slower & more variable (Wi-Fi, cellular networks)
INSTALLATION	More complex (requires cables, routers, switches)	Easier (just a router or access point) **Ad Hoc Mode** – Direct device-to-device communication (No central access point) **Infrastructure Mode** – Devices connect via a Wi-Fi router or access point
REALIABILITY	Highly stable (less interference) and More secure (physical access needed)	Less stable (affected by interference & distance) and Less secure (susceptible to hacking)
MOBILITY	Cannot provide mobile network and Limited (fixed location)	High (move freely within range)
COST	High (cables, routers, switches, labor)	Lower (just a router or access point)

HOW WIRELESS TECHNOLOGY WORKS

- **Wireless technology** allows devices to communicate without physical cables by using radio waves, infrared, or satellite signals to transmit data.

1. **Transmitter** – Sends signals (e.g., Wi-Fi router, cell tower, satellite).
2. **Receiver** – Receives signals (e.g., smartphone, laptop, GPS device).
3. **Medium** – The air (radio waves, infrared, microwave, etc.).
4. **Protocols** – Rules for communication (Wi-Fi, Bluetooth, 4G/5G, etc.).

TYPES OF TECHNOLOGIES

WI-FI (WIRELESS FIDELITY)

- Is a generic term that refers to the **IEEE 802.11** communications standard for Wireless Local Area Networks (WLANs).
- When Wi-Fi enabled device encounters a hotspot, the device can then connect to that network wirelessly.



TYPES OF TECHNOLOGIES (CONTINUES…)

BLUETOOTH

- Is a wireless technology for transferring data between two devices that are in close proximity with each other.



ADVANTAGES & DISADVANTAGES OF WIRELESS TECHNOLOGY

	ADVANTAGES (PROS)	DISADVANTAGES (CONS)
Mobility	Devices can connect anywhere within range	Signal weakens with distance & obstacles
Installation	No cables, easy & quick setup	Requires a router, access points, or mobile networks
Cost	No need for expensive cabling	Long-term costs (router upgrades, bandwidth fees) can add up
Speed	5G & Wi-Fi 6 offer fast speeds	Generally slower than wired connections, especially under heavy usage
Security	Can use encryption (WPA3, VPNs) for protection	Susceptible to hacking, unauthorized access, and signal interception

APPLICATIONS OF WIRELESS TECHNOLOGY

1. **Remote Control**

- Wireless technology enables remote operation of devices such as TVs, drones, garage doors, and industrial machinery.
- Infrared (IR) and radio frequency (RF) signals are commonly used for this purpose.

APPLICATIONS OF WIRELESS TECHNOLOGY (CONTINUES…)

2. **Item Tracking**

- Wireless tracking technologies like RFID (Radio-Frequency Identification) and GPS (Global Positioning System) help monitor the location of valuable assets, inventory, and personal belongings in real time.

APPLICATIONS OF WIRELESS TECHNOLOGY (CONTINUES…)

3. **Navigation And Location**

- The Global Positioning System (GPS) is a space-based Global Navigation Satellite System (GNSS) that provides location and time information.

APPLICATIONS OF WIRELESS TECHNOLOGY (CONTINUES…)

4. **Wireless Power Transfer**

- Is a technology that allows electrical energy to be transmitted from a power source to a device without using physical wires.
- For example, Smartphone and Electric Vehicle (EV) Wireless Charging.

APPLICATIONS OF WIRELESS TECHNOLOGY (CONTINUES…)

5. **Remote Monitoring**

- Wireless sensors and IoT (Internet of Things) devices help monitor health (wearable devices), industrial machines, and smart homes for security, temperature control, and energy management.

APPLICATIONS OF WIRELESS TECHNOLOGY (CONTINUES…)

6. **Risk Management**

- Wireless surveillance systems, fire alarms, and disaster monitoring networks help detect potential threats and mitigate risks in security, environmental protection, and industrial operations.

WIRELESS DATA NETWORKS

UNIT 2

INTRODUCTION

- We can say that **wireless networks** are those which are not connected by any type of cables and it uses different wireless connections between network nodes.
- These networks allow devices like smartphones, laptops, and IoT devices to connect and exchange data over a distance.

NEED OF WIRELESS DATA NETWORKS

	NEED
Mobility & Convenience	Users can stay connected **without being tied to a physical cable.** Ideal for **mobile devices,** allowing access to the internet anywhere (home, office, public spaces).
Scalability & Flexibility	Easy to expand **without adding new cables.**
Cost-Effective	Reduces costs related to **cabling, installation, and maintenance.**
Facilitates Public Wi-Fi & Hotspots	Used in **cafés, airports, schools, and shopping malls** to provide internet access.
Connectivity in Remote Areas	Provides internet access where **wired infrastructure is unavailable,** such as **rural areas, disaster zones, and mobile networks (3G, 4G, 5G, satellite internet).**

BENEFITS & DISADVANTAGES OF WIRELESS DATA NETWORKS

BENEFITS	DISADVANTAGES
Mobility & Convenience – Users can connect from anywhere without being tied to cables.	**Security Risks** – Wireless networks are more vulnerable to hacking, interception, and unauthorized access.
Cost-Effective – Reduces costs associated with cabling, installation, and maintenance.	**Slower Speeds** Compared to Wired Networks
	Limited Range – Wi-Fi signals have a limited coverage area.
	Interference – Signals can be weakened by walls

TYPES OF WIRELESS DATA NETWORKS

1. **Wireless Local Area Network (WLAN)**

- Used in homes, offices, and public places to connect multiple devices within a limited range.
- For example Wi-Fi.



- In a Local Area Network (LAN) a group of computers and other devices have a common wireless network within a distinct area.

TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

Types of WLANs

i. **Infrastructure** – Devices connect via a Wi-Fi router or access point.

ii. **Peer-to-Peer** or **Ad Hoc** – Direct device-to-device communication (No central access point).

iii. **Wireless Distribution System** – Wireless is expanded using multiple access points without the need for wired backbone to link them.

TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

2. **Wireless Metropolitan Area Network (WMAN)**

- Covers a city or large campus, providing high-speed wireless internet over long distances.
- This cover a wide area than Local Area Network (LAN) but less than the Wireless Wide Area Network (WWAN).
- For example, WiMAX.

TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

- In this wireless network type, access points are placed on the poles or on the buildings through the area.
- When users are connecting to their destination they are connected to the nearest access point.



TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

ADVANTAGES	DISADVANTAGES
Sharing of connections – In metropolitan area network it is a combination of two or more LAN networks	**Difficulty in managing** – It may be difficult to manage if number of lines are increased. This is due to the security purposes.
Security level – It has high security than WAN	
Cost – It is less expensive with good data efficiency	
High speed – The speed in this network is fast because MAN uses fiber optics	**Speed** – Of network may be slow due to high coverage

TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

3. **Wireless Personal Area Network (WPAN)**

- Covers a short range (a few meters e.g., 100) for connecting personal devices like phones, smartwatches, and wireless earbuds.
- For example, Bluetooth, Zigbee, UWB.



TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

BENEFITS	DISADVANTAGES
No need of data cables	Transferring of data may be slow
Data can be transfer between multiple devices at a time	It can only be used in personal area limited coverage
Not much costly	

TYPES OF WIRELESS DATA NETWORKS (CONTINUES…)

4. **Wireless Wide Area Network (WWAN)**

- It is different from the rest of the networks because it uses mobile telecommunications and other cellular technologies like 2G, 3G, 4G and 5G in transferring of data.
- It basically covers a large geographical area.



WIRELESS PERSONAL AREA NETWORK (WPAN)

UNIT 3

IEEE DEFINITION OF WPAN

- **Wireless Personal Area Networks (WPANs)** are used to convey information over short distances among a private, intimate group of participant devices.

WPAN TECHNOLOGIES

1. BLUETOOTH

- Is a cable-replacement technology that can be used to connect almost any device to any other device.
- It as a radio interface enabling electronic devices to communicate wirelessly via short range (**10 meters**) **ad-hoc radio connections**.
- Uses the radio range of **2.45 GHz.**
- Theoretical maximum **bandwidth** is 1 **Mb/s.**

WPAN TECHNOLOGIES (CONTINUES…)

Bluetooth Products

1. Personal Computers (PCs)
2. Printers
3. Cell Phones
4. Headsets & Speakers
5. Cameras



WPAN TECHNOLOGIES (CONTINUES…)

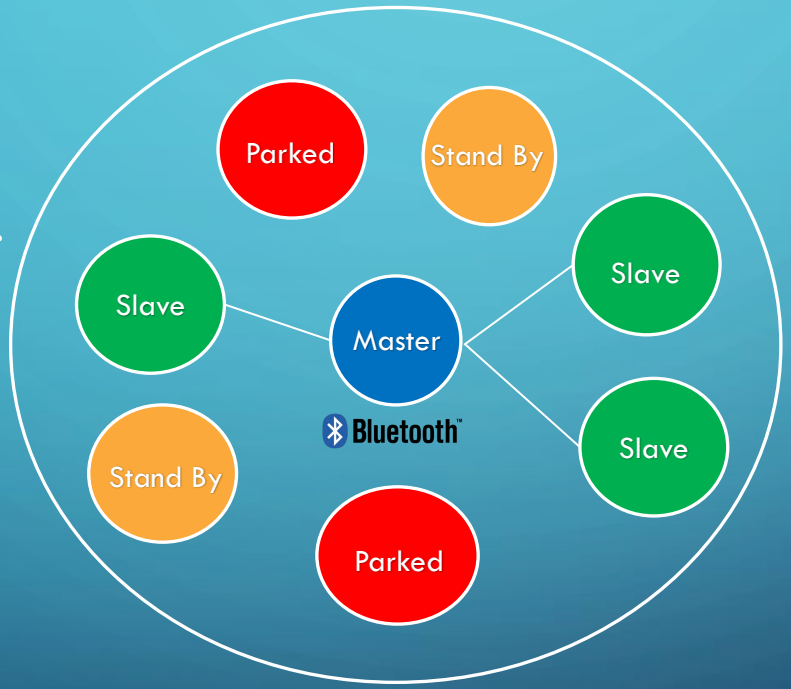
Bluetooth Architecture

i. Piconet

- Several Bluetooth devices can form an **ad hoc network** called a "**piconet**".
- Each piconet has one **master** and up to 7 simultaneous **slaves.**
- **Master –** device that initiates a data exchange.
- **Slave** – device that responds to the master.
- All devices in a piconet hop together, master gives slaves its **clock** and device **ID**.
- Non-piconet devices are in standby.

WPAN TECHNOLOGIES (CONTINUES…)

- A **Piconet** is a small Bluetooth network consisting of one master device and up to seven **active** slave devices.



- There can also be devices in a **non-active** state:

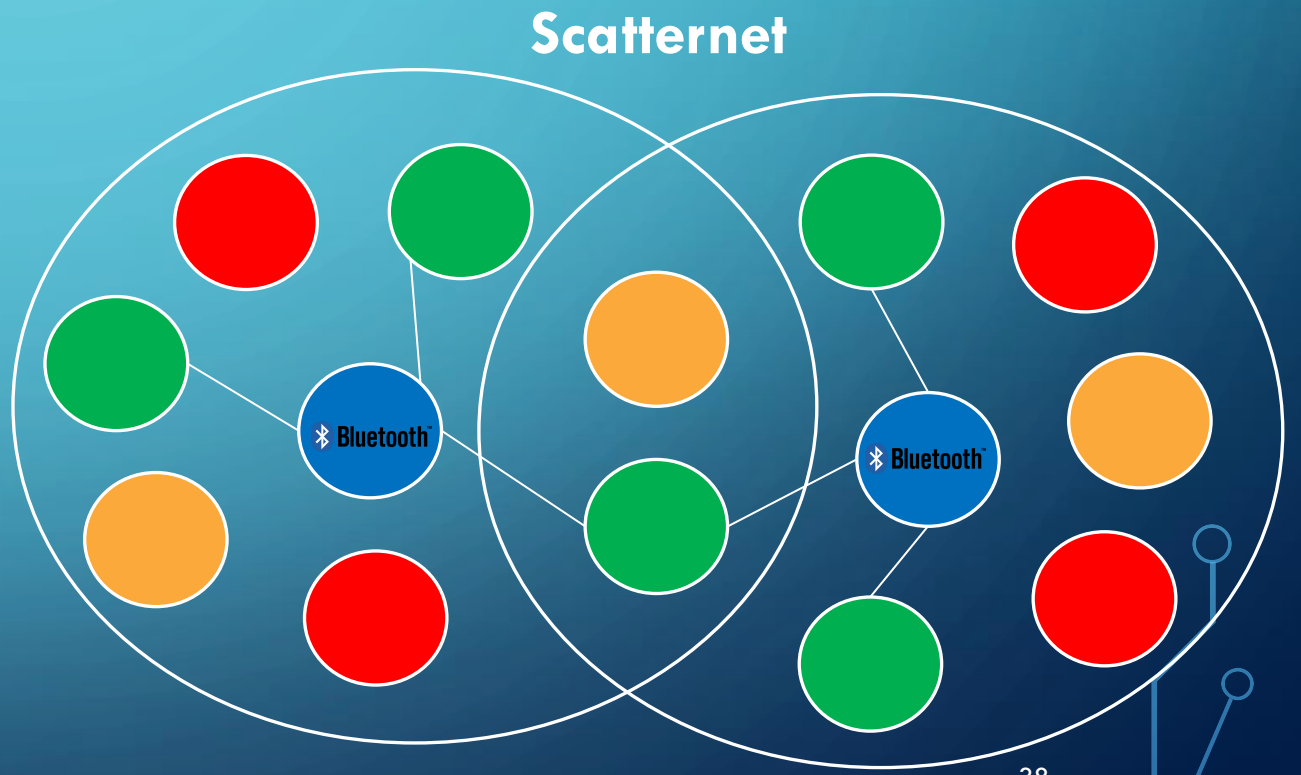
a) **Standby Mode** – A Bluetooth device is not connected to any piconet but is waiting for connection requests.

b) **Parked Mode** – A device is connected to the piconet but not actively participating in data exchange. Parked devices do not have an active address but can be reactivated by the master.

WPAN TECHNOLOGIES (CONTINUES…)

ii. **Scatternet**

- Is the linking of multiple piconets through the master or slave devices.
- Devices can be slave in one piconet and master of another.



WPAN TECHNOLOGIES (CONTINUES…)

Physical links

- Between master and slave(s), different types of links can be established.

i. **Synchronous Connection Oriented (SCO)**

- Support **symmetrical, circuit-switched, point-to-point connections.**
- Typically used for voice traffic.
- Data rate is 64 kbit/s.

WPAN TECHNOLOGIES (CONTINUES…)

ii. **Asynchronous Connection-Less (ACL)**

- Support **symmetrical** and **asymmetrical, packet-switched, point-to-multipoint connections**.
- Typically used for data transmission.
- Data rate is 433.9 kbit/s in symmetric or 723.2/57.6 kbit/s in asymmetric.

WPAN TECHNOLOGIES (CONTINUES…)

BLUETOOTH SECURITY	
Authentication	Ensures that devices communicating over Bluetooth are **legitimate** and **trusted.** Uses **pairing** and **mutual authentication** to **verify the identity** of devices before establishing a connection. **Authentication** is performed using a **shared secret key** generated during the pairing process.
Confidentiality	**Protects** Bluetooth communication from eavesdropping by encrypting transmitted data. Uses **symmetric encryption** (such as **E0 stream cipher**) to prevent unauthorized access.
Authorization	After authentication, the master device determines **which services** the slave device can access. For example, a Bluetooth headset may be allowed to access audio functions but **not** transfer files from your phone

WPAN TECHNOLOGIES (CONTINUES…)

2. **ZigBee**

- Is a technological standard designed for control and sensor networks.
- Based on the **IEEE 802.15.4** Standard.
- Created by the **ZigBee Alliance**.

WPAN TECHNOLOGIES (CONTINUES…)

Characteristics

i. Low cost
ii. Low power consumption
iii. Low data rate
iv. Scalability
v. Reliability
vi. Flexibility
vii. Relatively short transmission rate

WPAN TECHNOLOGIES (CONTINUES…)

ZIGBEE SECURITY	
Authentication	Ensures only trusted devices can join the network.
Encryption	Uses **AES-128 encryption** to protect data from eavesdropping. **CTR (Counter Mode)** – Used for **encryption**, turning plaintext into ciphertext using a counter value.
Integrity	Prevents message tampering using **Message Integrity Codes (MIC).** **CBC-MAC (Cipher Block Chaining – Message Authentication Code)** – Used for **data integrity**, ensuring messages are not altered.
Access Control	Devices must have an authorized key to communicate.

WPAN TECHNOLOGIES (CONTINUES…)

ZigBee Applications

i. **Home Automation** – Smart lighting and remote-controlled appliances.

ii. **Industrial Automation**

iii. **Healthcare & Medical Devices** – Patient monitoring systems and wearable health devices.

WPAN TECHNOLOGIES (CONTINUES…)

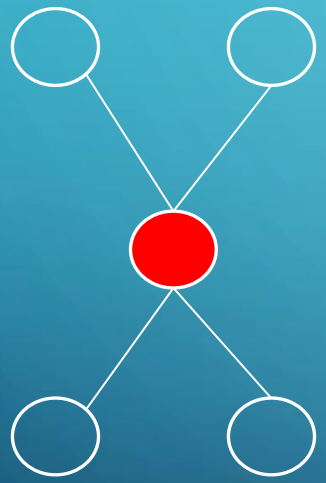
ZigBee Topologies

- A **topology** is the arrangement or structure of a network, defining how devices are interconnected and communicate with each other.

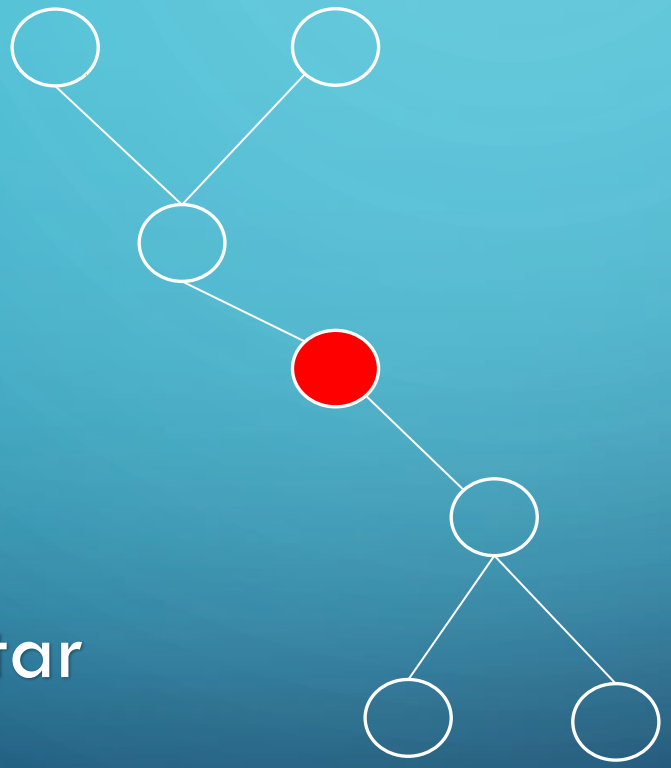
1. **Star** – is a network configuration where all devices are connected to a central hub or switch, which manages data transmission between them.
2. **Mesh** – is a network configuration where each device is interconnected with multiple other devices, ensuring redundancy and reliable communication.
3. **Cluster Tree** – is a network structure where devices are organized into hierarchical clusters, with a central coordinator managing communication and routing data between clusters.

WPAN TECHNOLOGIES (CONTINUES…)

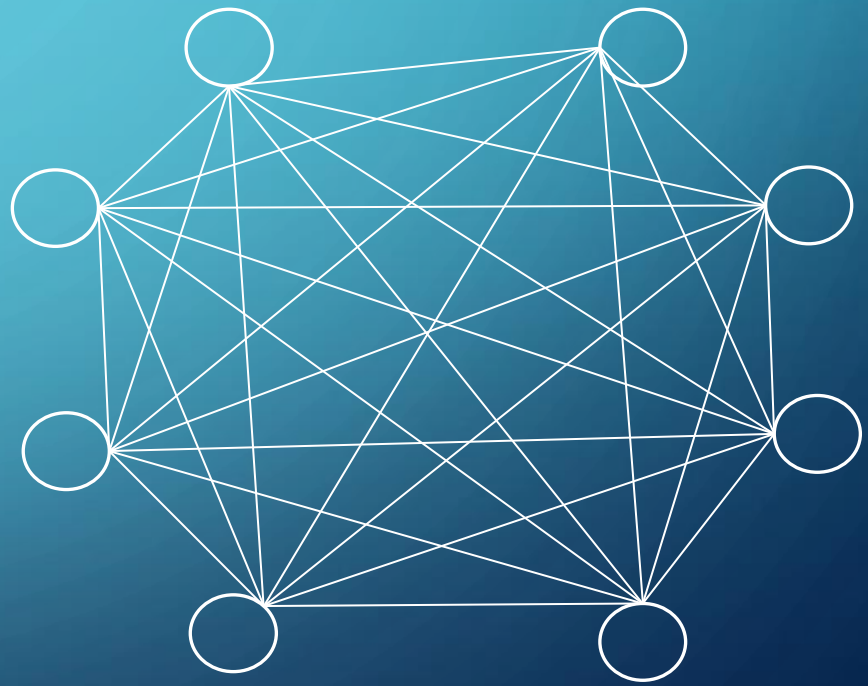
ZigBee Topologies



Star



Cluster Tree



Mesh

WPAN TECHNOLOGIES (CONTINUES…)

	ADVANTAGES	DISADVANTAGES
Star	Easy to install and manage. A failure in one device does not affect the entire network.	Failure of the central hub disrupts the network. More cabling is required, increasing cost.
Mesh	Provides redundancy and fault tolerance.	Expensive to implement due to high cabling and hardware costs. Complex configuration and maintenance.
Cluster Tree	Scalable and supports large networks. Efficient power consumption, making it suitable for wireless sensor networks.	Failure of a parent node can isolate entire clusters.

WPAN TECHNOLOGIES (CONTINUES…)

ZigBee and Bluetooth Comparison

	BLUETOOTH	ZIGBEE
Power Profile	Days	Years
Complexity	Complex	Simple
Nodes	7	64000
Latency	10 seconds	30ms – 1s
Range	10m	70m – 300m
Extendibility	No	Yes
Data Rate	1Mb/s	250kb/s
Security	64-bit, 128-bit	128-bit AES

WPAN TECHNOLOGIES (CONTINUES…)

3. **Ultra Wide Band (UWB)**

- Is a technology for short range Wireless radio communications.
- It as a bandwidth more than 25% of a center frequency or more than 1.5 GHz.

WIRELESS TRANSMISSION MEDIA

UNIT 4

WIRELESS TRANSMISSION

- **Wireless transmission** is a form of unguided media.
- Wireless signals are spread over in the air and are received and interpreted by appropriate antennas.
- Some **generation of wireless communication** include;

1. 1st Generation (1G)
2. 2nd Generation (2G)
3. 3rd Generation (3G)
4. 4th Generation (4G)
5. 5th Generation (5G)

TYPES OF WIRELESS TRANSMISSION

1. **Radio Wave**

- Is an electromagnetic phenomenon which travels through space and through many non-magnetic materials.
- They range from **3 KHz to 3000 GHz** and also **classified** as –

i. **Low Frequency** – 30 to 300 KHz.

ii. **Medium Frequency** – 300 to 3000 KHz.

iii. **High Frequency** – 300 to 3000 GHz.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…) ADVANTAGES OF RADIO WAVES

i. **Speed** – electromagnetic waves of any kind are the fastest things in the universe.
ii. **Penetration Power** – can pass through obstacles.
iii. **Low Power Consumption** – efficient for wireless devices.
iv. **Cost-Effective** – low infrastructure costs.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…) DISADVANTAGES OF RADIO WAVE

TYPES OF WIRELESS TRANSMISSION (CONTINUES…) CHARACTERISTICS OF A RADIO WAVE

i. Radio wave is a method of transmitting intelligence from one location to another by means of **electromagnetic** radiation.

ii. Radio waves can **travel through long distances** and **penetrate through materials** unlike light and heat waves.

iii. Radio waves are **invisible**.

iv. Radio waves are normally **not felt**.

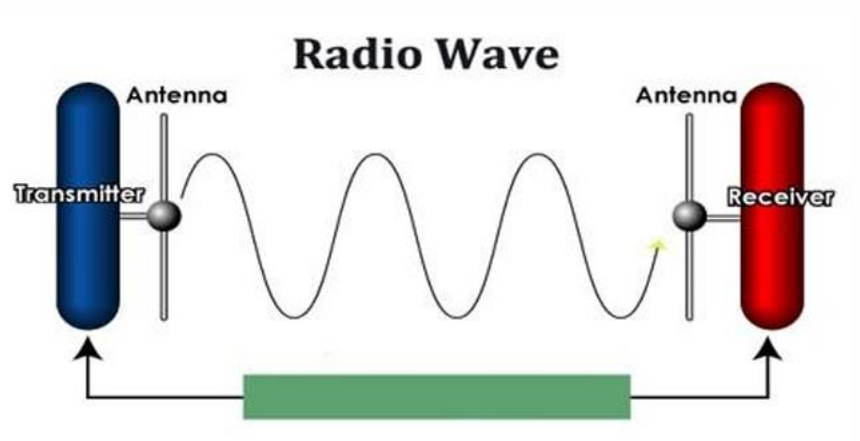
v. Radio waves have **frequencies** as high as 300 GHz to as low as 30 Hz.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…) APPLICATION OF RADIO WAVE

i. Radio waves carry signal for **AM** and **FM radio**.

ii. It is highly use in cellular **telephones** and **Television**.

iii. Radio waves are useful for **Police** and **Radar**.



TYPES OF WIRELESS TRANSMISSION (CONTINUES…)

2. Microwave

- Travels in **straight lines**, both sender and receiver must be aligned to be strictly in line-of-sight.
- Microwaves can have **wavelength** ranging from **1 mm – 1 meter** and **frequency** ranging from **300 MHz to 300 GHz**.
- Microwave transmission depends highly upon the **weather conditions** and the frequency it is using.
- Microwaves are used for **unicast communication** such as cellular telephones, satellite networks and WLANs.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…)
TYPES OF MICROWAVE

a) **Terrestrial Microwave**

- Is a wireless microwave **networking technology** that uses line-of sight communications between pairs of **Earth-based transmitters and receivers** to relay information.
- It is the most common form of long-distance communication.
- Its **characteristics** include –

i. Fewer Repeaters but requires **line of sight transmission**.

ii. Main **source of loss** is attenuation caused mostly by **distance, rainfall** and **interference.**

iii. Common **frequencies** used for transmission are in the range **1 to 40 GHz**.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…)
TYPES OF MICROWAVE (CONTINUES…)

b) Satellite Microwave

- Uses **satellites** for broadcasting and receiving of signals.
- These systems need satellites which are in the geostationary orbit which is **3600 km** above the earth.
- Its **advantages** include –

i. No cables needed

ii. Multiple channels available

iii. Wide bandwidth

TYPES OF WIRELESS TRANSMISSION (CONTINUES…)
TYPES OF MICROWAVE (CONTINUES…)

- Its **disadvantages** include –

i. **Expensive** towers and repeaters

ii. **Subject to interference** such as passing airplanes and rain

iii. **Line of sight** requirement

- Its major **characteristic** is the optimum **frequency range** for satellite transmission is **1 to 10 GHz**.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…)
TYPES OF MICROWAVE (CONTINUES…)
APPLICATIONS OF SATELLITE MICROWAVE

i. Television distribution

ii. Long distance telephone transmission

iii. Global Positioning System (GPS)

TYPES OF WIRELESS TRANSMISSION (CONTINUES…)

3. Infrared (IR)

- **Involves sending light signals** at a frequency between visible light and radio waves.
- **Commonly used in TV remote controls,** now is used to provide LAN connections.
- It is a **line-of-sight** transmission and has a maximum coverage of **30 to 80 feet.**

TYPES OF WIRELESS TRANSMISSION (CONTINUES…) CHARACTERISTICS OF IR

i. Sends signals through **infrared light**.

ii. Sending a signal in a **straight line**.

iii. **Cannot penetrate** through the surface of the object.

iv. It is generally **used for short distance** transmission such as a TV remote.

v. It does not require an antenna to operate.

TYPES OF WIRELESS TRANSMISSION (CONTINUES…) APPLICATIONS OF IR

i. Remote controls for television.

ii. Wireless LANs.

iii. Intrusion Detection Systems (IDS).

WIRELESS SECURITY THREATS

UNIT 5

SECURITY TREAT & BREACH

- A **threat** is any potential danger that can exploit a vulnerability in a system or network to cause harm, damage, or unauthorized access to data, resources, or services.
- In other words, it is anything that can compromise the confidentiality, integrity, or availability of information or systems.
- A **security breach** is an incident where unauthorized individuals gain access to a system, network, or data, violating the system's security policies.

SECURITY TREAT & BREACH (CONTINUES…)

Aspect	Threat	Breach
Timing	Before the attack	After the attack
Action	Potential attack	Successful attack
Impact	No damage yet	Actual damage/loss
Examples	Malware, phishing attempts, unpatched software	Data theft, system compromise, information leak

SECURITY THREATS

1. Eavesdropping

- Is the act of secretly or stealthily listening to the private conversation or communications of others without their consent in order to gather information.
- Eavesdropping is especially dangerous because wireless signals travel through the air and can be intercepted without physical access.
- For example, using Wireshark to capture data from users connected to the public Wi-Fi to steal unencrypted data such as passwords, emails, or credit card numbers.

SECURITY THREATS (CONTINUES…)

TYPES OF EAVESDROPPING

i. **Passive** – Just listens to data without altering it.

ii. **Active** – May inject queries or interfere to gather more data.

PREVENTION OF EAVESDROPPING

i. **Encryption** - WPA3 for Wi-Fi, SSL/TLS for web.

ii. **VPNs**

iii. Avoid using public Wi-Fi for sensitive tasks.

SECURITY THREATS (CONTINUES…)

2. Communication Jamming

- Refers to the deliberate interference with wireless communications.
- It is a form of Denial of Service (DoS) attack specifically aimed at wireless communication channels.

SECURITY THREATS (CONTINUES…)

Types of Jamming

i. **Spot** – Used to block a single communication channel.

ii. **Sweep** – Moves a jamming signal across multiple frequencies.

iii. **Barrage** – Jams a wide range of frequencies at once.

iv. **Deceptive** - Sends fake signals to mislead or confuse receivers.

SECURITY THREATS (CONTINUES…)

EFFECTS AND RISKS OF JAMMING

i. Loss of communication
ii. Interruption of Wi-Fi
iii. Loss of service on commercial networks

COUNTERMEASURES OF JAMMING

i. Frequency hopping
ii. Spread spectrum
iii. Shielding and filtering
iv. Signal detection

SECURITY THREATS (CONTINUES…)

3. Radio Frequency (RF) Interference

- RFI also known as **electromagnetic interference (EMI)** is when external signals mess with the wireless signals.
- RF interference can be intentional (like jamming) or unintentional (accidental).

SECURITY THREATS (CONTINUES…)

SOURCES OF RF INTERFERENCE

Natural	Man-made
Lightning	Cell phones and towers
Solar flares	Wi-Fi routers
Cosmic radiation	Radios and TVs

EFFECTS OF RF INTERFERENCE

i. Dropped calls in mobile networks

ii. Poor Wi-Fi performance

iii. Disrupted broadcast signals (TV, radio)

iv. Errors in sensitive instruments (e.g., medical equipment)

SECURITY THREATS (CONTINUES…)

Types of RF Interference

i. **Co-channel interference** – Occurs when two transmitters use the same frequency channel.

ii. **Adjacent-channel interference** – Happens when signals from nearby channels overlap.

iii. **Intermodulation interference** – Caused when multiple RF signals mix together and produce unwanted frequencies.

iv. **Harmonic interference** – Comes from signals that are multiples of a base frequency, unintentionally picked up by other devices.

SECURITY THREATS (CONTINUES…)

Prevention & Mitigation of RFI

i. Shielding & Filtering

ii. Spread Spectrum

SECURITY THREATS (CONTINUES…)

4. **Structured Query Language Injection (SQLi)**

- Is a code injection attack where a malicious user inserts or "injects" harmful SQL code into input fields or parameters of an application allowing attackers to interfere with the queries an application makes to its database.
- It's effects include;

i. Unauthorized access to sensitive data

ii. Modify or delete data in the database

iii. Control the entire database by administrative privilege escalation

SECURITY THREATS (CONTINUES…)

How to Prevent SQL Injection

i. User input validation

ii. Least privilege principle

iii. Regularly update and patch software

SECURITY THREATS (CONTINUES…)

5. Denial of Service (DoS)

- Is a malicious attempt to make a computer, server, or network resource unavailable to its intended users either temporary or permanently by overwhelming it with a flood of fake traffic or requests.

SECURITY THREATS (CONTINUES…)

Types of DoS Attacks

i. **Volume-Based Attacks** – Flood the target with huge amounts of traffic (e.g., UDP Flood, ICMP Flood)

ii. **Protocol Attacks** – Exploit weaknesses in network protocols (e.g. Ping of Death) so as to exhaust server resources like CPU or memory

iii. **Application Layer Attacks** – Target specific apps or services (e.g., HTTP Flood) so as to exhaust the application's resources

- Others

a) Buffer Overflow (Stack And Heap)

b) SYN Flood/ Half-open attack

c) Teardrop

d) Smurf

SECURITY THREATS (CONTINUES…)

6. **Spoofing**

- Is a cyberattack where an attacker pretends to be someone else by faking data to trick a system or user into trusting them.
- This can be done on;

a) IP Address (IP Spoofing)
b) Email Address (Email Spoofing)
c) MAC Address (MAC Spoofing)
d) Caller ID (Caller ID Spoofing)
e) Website/URL (Website Spoofing)

SECURITY THREATS (CONTINUES…)

REASONS FOR SPOOFING

i. To bypass security controls

ii. To steal data or credentials

iii. To hide their true identity

HOW TO PROTECT AGAINST SPOOFING

i. **Encryption** – HTTPS and VPNs

ii. **Email authentication** – SPF, DKIM and DMARC

iii. **Network security measures** – Firewalls, honey pots, ACLs, IPS and IDS

iv. **Awareness**

TRAFFIC ANALYSIS

UNIT 6

TRAFFIC ANALYSIS

- Is the process of monitoring, capturing, and examining wireless network traffic to gather information about communication patterns, even if the actual content is encrypted.
- It analyzes metadata and observable characteristics of network traffic flows, such as Source & Destination Addresses/MACs and Packet Sizes & Lengths.
- Unlike eavesdropping which aims to decrypt content, TA infers, meaning from the context and pattern of the communication, it gather information even when content is encrypted.

WHY IS TA A CRITICAL THREAT IN WIRELESS NETWORKS?

1. **Inherent Broadcast Medium/Security** – Wireless signals propagate through the air, making them inherently easier to intercept.
2. **Metadata Exposure** – While encryption (WPA2/WPA3) protects payload content, fundamental metadata (source, destination, size, timing) is often still visible in unencrypted frame headers or easily observable patterns.
3. **Defeating Encryption** – TA bypasses strong encryption algorithms.
4. **Stealth** – Passive TA is extremely difficult to detect.

TYPES OF TRAFFIC ANALYSIS ATTACKS

PASSIVE ATTACKS

- The attacker only listens and doesn't inject traffic or disrupt communications.
- It can be done through **monitoring, pattern recognition, correlation attacks** and **social network analysis.**

ACTIVE ATTACKS

- The attaker inject traffic or disrupt communications.
- It can be done through **packet injection, traffic generation** and **selective jamming.**

COMMON TA TECHNIQUES & TOOLS

TECHNIQUES

1. Flow Analysis
2. Statistical Analysis
3. Time-Series Analysis
4. Website Fingerprinting
5. Entropy Analysis
6. Machine Learning (ML)

TOOLS

1. Wireshark
2. tcpdump
3. Aircrack-ng suite

WIRELESS SECURITY STANDARDS

UNIT 7

WIRELESS SECURITY STANDARDS

- Wireless security standards are protocols designed to protect wireless networks (Wi-Fi) from unauthorized access, data breaches, and interference.

1. **Wired Equivalent Privacy (WEP)**

- Introduced in 1997 as part of the original Institute of **Electrical and Electronics Engineers (IEEE) 802.11** standard.
- The goal was to provide privacy similar to that of wired networks.
- It uses **Rivest Cipher 4 (RC4)** stream cipher for encryption and an **Initialization Vector (IV)** 24-bit field added to it.
- Key length 64-bit (40-bit key + 24-bit IV) or 128-bit (104-bit key + 24-bit IV).
- Authentication methods include **Open System** (any device can authenticate) and **Shared Key** (devices authenticate using a shared secret)

WIRELESS SECURITY STANDARDS (CONTINUES…)

WEAKNESSES OF WEP

i. IVs are too short and frequently reused

ii. RC4 is vulnerable to statistical attacks

iii. Key reuse makes WEP vulnerable to replay attacks

REAL-WORLD ATTACK

- Tools like Aircrack-ng can capture encrypted packets and use them to recover WEP keys.
- Declared insecure by the Wi-Fi Alliance in 2004.

WIRELESS SECURITY STANDARDS (CONTINUES…)

2. Wi-Fi Protected Access (WAP)

- Introduced in 2003 by the **Wi-Fi Alliance**.
- Interim solution to replace WEP until IEEE 802.11i was finalized.
- It uses a **Temporal Key Integrity Protocol (TKIP)** which dynamically generates new keys for every packet and appends a **Message Integrity Check (MIC)** to detect tampering.
- Still uses RC4 for encryption.

WIRELESS SECURITY STANDARDS (CONTINUES…)

WPA MODES

i. **WPA-Pre-Shared Key (PSK)** – For home users.

ii. **WPA-Enterprise** – Uses IEEE 802.1X authentication and a Remote Authentication Dial-In User Service (RADIUS) server.

WEAKNESSES & REAL-WORLD ATTACK

i. TKIP still relies on RC4, inheriting many of its weaknesses

ii. Vulnerable to replay attacks and MIC key recovery

- **Beck-Tews** attack allows packet injection by exploiting TKIP flaws

WIRELESS SECURITY STANDARDS (CONTINUES…)

3. **Wi-Fi Protected Access II (WPA2)**

- Released in 2004, based on the **IEEE 802.11i** amendment.
- Made mandatory for Wi-Fi certified devices by 2006.
- It used **Advanced Encryption Standard (AES)** and **Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP).**
- Provides both **encryption** and **message integrity**.
- Uses a **4-way handshake** to establish encryption keys.
- Authentication Modes include **WPA2-PSK** and **WPA2-Enterprise.**
- **Key Reinstallation Attack (KRACK**) vulnerability of 2017 exploits flaws in the 4-way handshake process to reinstall already-used keys.

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WIRELESS SECURITY STANDARDS (CONTINUES…)

4. Wi-Fi Protected Access III (WPA3)

- Introduced in 2018 to overcome WPA2 limitations.
- Replaces WPA2's PSK with **Simultaneous Authentication of Equals (SAE)** which is based on **Password-Authenticated Key Exchange (PAKE)**, providing resistance to dictionary attacks.
- Past session keys remain secure even if a current key is compromised (**Forward Secrecy**).

WIRELESS SECURITY STANDARDS (CONTINUES…)

ADDITIONAL FEATURES

i. **Opportunistic Wireless Encryption (OWE)** – Provides encrypted communication in open networks

ii. **Wi-Fi Easy Connect** – Uses **Device Provisioning Protocol (DPP)** to connect IoT (Internet of Things) devices securely via QR codes

VULNERABILITY

i. **Dragonblood attacks (2019)** – Exploited implementation flaws in the SAE protocol, now mitigated in updates

WIRELESS SECURITY STANDARDS (CONTINUES…)

5. **Wireless LAN Security (IEEE 802.11i)**

- Finalized in 2004 and introduced the **Robust Security Network (RSN)** concept.
- It's **features** included;

i. AES-CCMP

ii. 802.1X

iii. Key management, authentication, and data integrity

WIRELESS SECURITY STANDARDS (CONTINUES…)

6. **Port-Based Network Access Control (IEEE 802.1X)**

- Provides a framework for authenticating and authorizing devices on a network.
- It's **components** include;

i. **Supplicant** - The client device requesting access

ii. **Authenticator** – Network device (e.g., Access Point or Switch)

iii. **Authentication Server** - Usually a RADIUS server

WIRELESS SECURITY STANDARDS (CONTINUES…)

Extensible Authentication Protocol (EAP) Variants Used in IEEE 802.1X

i. **EAP-Transport Layer Security (TLS)** – Uses client and server certificates
ii. **Protected EAP (PEAP)** – Encapsulates EAP inside a TLS tunnel
iii. **EAP-Tunneled Transport Layer Security (TTLS)**

4-WAY HANDSHAKE

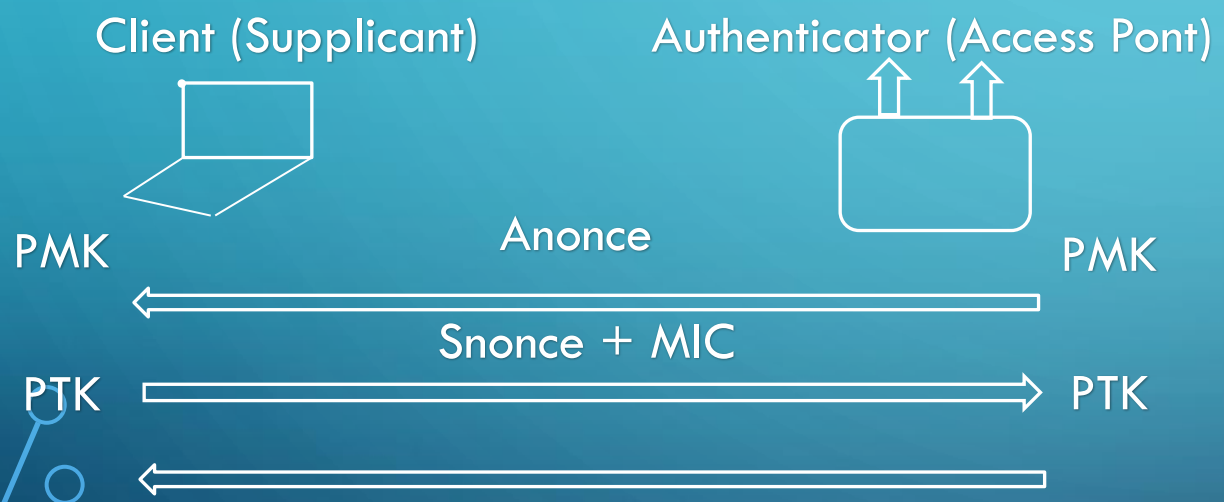
- Is a process used in wireless security protocols, particularly Wi-Fi Protected Access protocols, to establish a secure connection between a Wi-Fi client (like your laptop or phone) and a Wi-Fi access point (router).
- Its **purpose** is to;

i. To confirm both the client and access point know the Pre-Shared Key (PSK) or use 802.1X authentication.

ii. To generate and exchange encryption keys (called Pairwise Transient Keys or PTKs).

iii. To protect against replay and man-in-the-middle attacks.

4-WAY HANDSHAKE (CONTINUES…)



WIRELESS TRANSPORT LAYER SECURITY(WTLS)

UNIT 8

WIRELESS TRANSPORT LAYER SECURITY (WTLS)

- **WTLS** is a security protocol specifically designed for wireless networks, particularly for devices using the **Wireless Application Protocol (WAP)**.
- **WTLS** is based on **TLS (Transport Layer Security)**, but it's optimized for wireless environments where bandwidth, processing power, and memory are limited.
- Below are some reasons **why we need WTLS**;

i. Communication

ii. Using Internet

TRANSPORT LAYER SECURITY (TLS)

- **TLS (Transport Layer Security)** is a cryptographic protocol designed to provide secure communication over a computer network, such as the internet.
- Transport Layer Security (TLS) is a slightly modified version of **SSL version 3**.
- **WTLS Functions** include –

i. Integrity

ii. Authentication

iii. Privacy

WTLS SPECIFICATIONS

1. **Full Handshaking**

- A Full Handshake is used when –

i. The client and server **have no prior session** or session data (i.e., **first-time connection**).

ii. The session must establish **new keys, perform certificate validation,** and **negotiate security parameters from scratch.**

WTLS SPECIFICATIONS (CONTINUES…)

- **Key Features of a full handshake** include –

i. Exchange of client and server **certificates** (if required).

ii. Generation of new **session keys**.

iii. **Longer time and processing** due to complete setup.

iv. **More secure**, especially when no trust has been previously established.

WTLS SPECIFICATIONS (CONTINUES…)

2. **Abbreviated Handshaking**

- An Abbreviated Handshake is used when –

i. The client and server have **previously communicated**.

ii. They **use cached session data** to resume a previous secure connection.

- Its **key features** include –

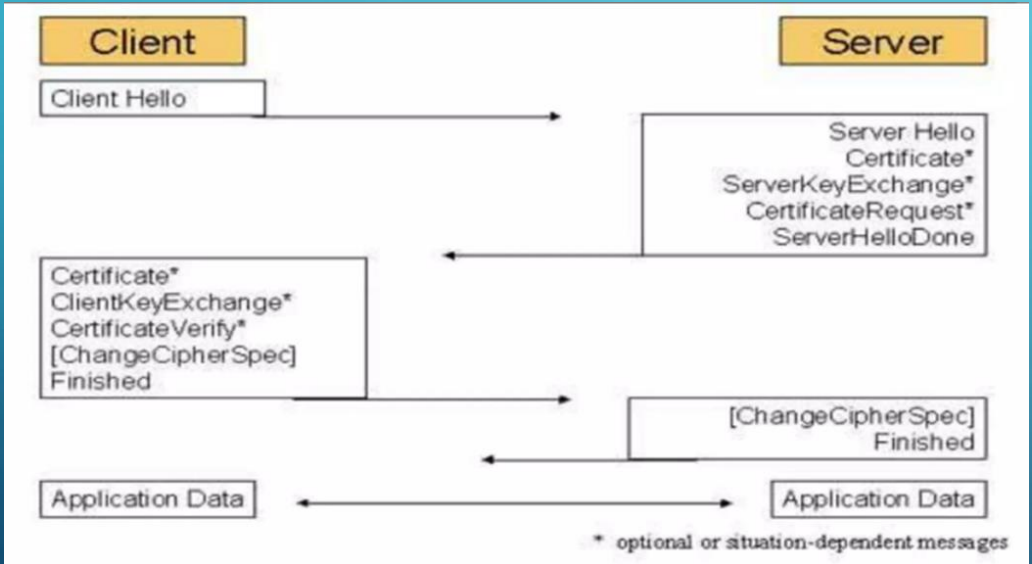
i. **Faster connection** setup.

ii. **Reuses previous session keys** or resumption secrets.

iii. **Saves on CPU**, memory, and bandwidth.

FULL VS. ABBREVIATED HANDSHAKE

FULL



ABBREVIATED

